Adaptive Memory Evolution: Controlling Stability–Plasticity in Agent Memory

Anonymous Authors

Abstract

Persistent memory must preserve useful history while responding to revised facts and changing demand. *Adaptive Memory Evolution* (AME) pairs content with evolving importance, consistency, usage, and temporal states. Frozen controller weights and rate interventions change behavior without retraining. Experiments show benefits under small capacity and interference, but not uniform superiority. Across four environments and 84 rate pairs, 27 exhibit opposing retention and adaptation changes significant after multiplicity correction. Replacement controls support attribution beyond counts, not full causal identification. Semantic transfer exposes stale retrieval despite preserved evidence. These findings characterize conditional stability–plasticity control.

1 Introduction

An assistant must retain enduring interests, replace outdated instructions, and recover dormant preferences. With finite storage, historical usefulness can become a liability after a context shift: stability and plasticity are distinct requirements.

Existing systems support retrieval, reflection, consolidation, tiered storage, and updates [1–4]. We isolate how per-item state changes alter the retention–adaptation balance, separately from architecture and replacement volume. Aggregate accuracy can hide dormant-knowledge loss and obsolete evidence.

With controller weights frozen, we test rate interventions on per-item state. Our contributions are:

1. A formulation separating immutable content, evolving state, addressing, and bounded residency, with distinct stability and plasticity objectives.
2. A hybrid evolution mechanism with explicit importance, consistency, and usage rates, event updates, and a common retention score.
3. A controlled characterization of trade-offs, replacement attribution, component roles, long horizons, and transfer failures using frozen streams and auditable trajectories.

Code: github.com/Nanoer-Wang/adaptive-memory-evolution.

2 Related Work

Agent memory systems. Generative Agents combine a memory stream with relevance, recency, importance, and reflection [1]. MemoryBank explicitly reinforces and forgets memories over time [3]; MemGPT manages memory across storage tiers [2]. A-MEM updates historical memory attributes and connections as new information arrives [4]. MemoryOS uses hierarchical updating [5], Mem0 extracts and consolidates conversational evidence [6], and AgeMem learns memory operations within the agent policy [7]. Evolving memory is therefore not unprecedented. We contribute a low-dimensional formulation and controlled characterization. Appendix I contrasts mechanisms and protocols; published scores are not matched-budget comparisons with ours.

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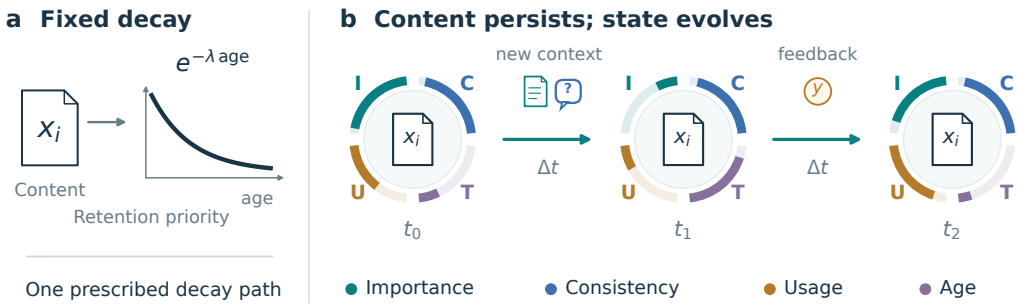


Figure 1: Immutable content, evolving state. Fixed decay prescribes a retention path; colored arcs depict qualitative state changes, not measured or normalized values.

Continual learning. Continual learning studies acquisition from changing streams without destructive forgetting [8]. For example, elastic weight consolidation constrains changes to parameters important for previous tasks [9]. We share the concern with stability and plasticity, but our controlled variable is an external record's retention state, not the agent's network weights. The head's weights remain frozen during evaluation. Stable-fact retention is not equivalent to preventing catastrophic forgetting in a learned representation.

Neural external memory. Neural Turing Machines and differentiable neural computers learn interactions with external memory through differentiable controllers [10, 11]. We store content explicitly; a 123-parameter affine head sets bounded insertion and importance coefficients and a positive rate. Closed-form integration is a convenient realization of the state equations. Its comparison with event-only and numerical integration variants tests the role of the dynamics without claiming a liquid neural network advantage.

Cognitive models and evaluation. Memory accessibility depending on the history of environmental demand has a long tradition [12]; strengthening and forgetting motivate our design without establishing cognitive fidelity. LongMemEval and LoCoMo expose temporal reasoning, knowledge updates, and multi-session recall challenges [13, 14]. MemoryBench evaluates memory and feedback-driven continual learning [15], while MemOps provides lifecycle operations and grounded update traces [16]. Our small transfer subsets and controlled public-text recombinations are explicitly distinct from official full-benchmark evaluations.

3 Problem: Retention under Changing Demand

Consider a timestamped stream $e_1, \ldots, e_N$ containing observations, context cues, queries, and post-answer feedback. At time t, a policy maintains a resident set $\mathcal{M}_t$ of at most K content records. Observations may introduce new facts or revised versions; a query asks for the currently valid value. A policy observes only the causal prefix. Stable/volatile labels and reference answers belong to the evaluator, not the policy.

Represent record i as $m_i(t) = (x_i, z_i(t))$, with immutable versioned content x_i and $z_i = (I_i, C_i, U_i, T_i)$. An addressing interface maps observations and queries to candidate identities. In structured experiments, identities and versions are supplied in the observed stream; semantic experiments infer them from resident embeddings. The latter is a substantially harder interface, not a change in the state equations.

We characterize a vector of outcomes: overall accuracy, stable retention, obsolete-answer error, and post-shift recovery. For a fixed environment $\mathcal{E}$, the sampled operating set is

$$\mathcal{O}_{\mathcal{E}} = \{(\mathrm{SR}(\pi_{\boldsymbol{\alpha}}), \mathrm{AS}(\pi_{\boldsymbol{\alpha}})) : \boldsymbol{\alpha} \in \mathcal{G}\}. \tag{1}$$

Controllability here means that a prespecified rate intervention can change the balance of these endpoints. It does not mean classical state-space controllability, monotonic performance, or an optimal frontier over all possible policies. Content capacity also differs from a total byte budget: auxiliary identity and query-history metadata remain outside the resident slots.

4 Adaptive Memory Evolution

4.1 Causal flow and event updates

At event j, the previous event's coefficients and context first evolve existing states through $\Delta_j = t_j - t_{j-1}$. Only then is the new event consumed. Let $q_i \in [0,1]$ denote held demand affinity and $v_i \in \{0,1\}$ indicate whether record i is the latest *observed* version of its addressed fact. Between events,

$$\begin{aligned} \dot{I}_i &= \alpha_I k(0.5 + wq_i - I_i), & \dot{C}_i &= 4\alpha_C k(v_i - C_i), \\ \dot{U}_i &= -\alpha_U k U_i/20, & \dot{T}_i &= 1. \end{aligned} \tag{2}$$

Importance I_i relaxes toward current demand; consistency C_i relaxes toward observed version status. Neither is an oracle of future utility or calibrated confidence. Usage U_i decays to reduce inertia from past successful use. Temporal state T_i ages between resets. For fixed held inputs, the update is exact:

$$I_i' = b_i + (I_i - b_i)e^{-\alpha_I k\Delta}, \quad C_i' = v_i + (C_i - v_i)e^{-4\alpha_C k\Delta}, \quad U_i' = U_i e^{-\alpha_U k\Delta/20}, \quad T_i' = T_i + \Delta, \tag{3}$$

where $b_i = 0.5 + wq_i$. Zero strength skips its corresponding flow exactly. It does not disable observation/feedback jumps or the temporal clock.

The coefficient head maps a 32-dimensional content projection and eight causal features to three outputs $r \in \mathbb{R}^3$: insertion gate $g = \sigma(r_1)$, importance amplitude $w = 2\sigma(r_2)$, and $k = (\mathrm{softplus}(r_3) + 10^{-4})/30$. This affine head has 123 parameters, pretrained in a separate earlier development stage and frozen for every reported validation batch (Appendix B). There is no new optimization for rate sweeps. The rule control fixes $g = 2/3, w = 1.5, k = 0.3$.

A newly observed version receives $(I, C, U, T) = (gw, 1, 0, 0)$; shared bootstrap records have $I = 1$. Reobserving a resident version applies $I \leftarrow \min(I + gw, 20)$, $C \leftarrow 1$, and $T \leftarrow 0$. After retrieval, delivered feedback $y \in [0,1]$ updates the retrieved record by

$$C \leftarrow 0.8C + 0.2y, \qquad U \leftarrow U + y, \qquad T \leftarrow 0. \tag{4}$$

Absent records receive no update. Thus the system is hybrid: both elapsed time and discrete evidence matter.

4.2 Retention, addressing, and interpretation

An arriving candidate is temporarily added to the resident set. If this causes overflow, the lowest-scoring candidate or resident is removed, using

$$S_i = \log\max(I_i, 10^{-8}) + \log\max(C_i, 0.05) + \log(1 + U_i) - 0.05\log(1 + T_i). \tag{5}$$

Away from the floors, this is $\log I_i + \log C_i + \log(1+U_i) - 0.05\log(1+T_i)$. The floors are part of the implementation. Rejecting a write is distinct from evicting an existing resident. Version metadata can survive eviction, but evicted content cannot be retrieved. Structured retrieval chooses the newest resident version of the requested key; Equation 5 controls availability, not answer ranking.

The flow has a useful local guarantee: under held inputs, I and C remain between their starting states and targets, while U contracts and T increases. A proof and the assumptions needed to bound repeated usage increments appear in Appendix A. This is a property of state evolution, not a guarantee of task accuracy: hard selection can switch discontinuously when score order changes.

5 Experimental Design and Results

5.1 Protocol and measurement

Structured streams recombine public bAbI facts [17] with independently seeded observation and query schedules. There are four stable facts and two groups of four revisable facts; one group is currently demanded. Observations, queries, and feedback interleave. Context switches and observed revisions are explicit. These are controlled management tasks, not the official bAbI reasoning benchmark. Content templates recur across worlds and earlier development data; new schedule seeds do not establish content-domain generalization.

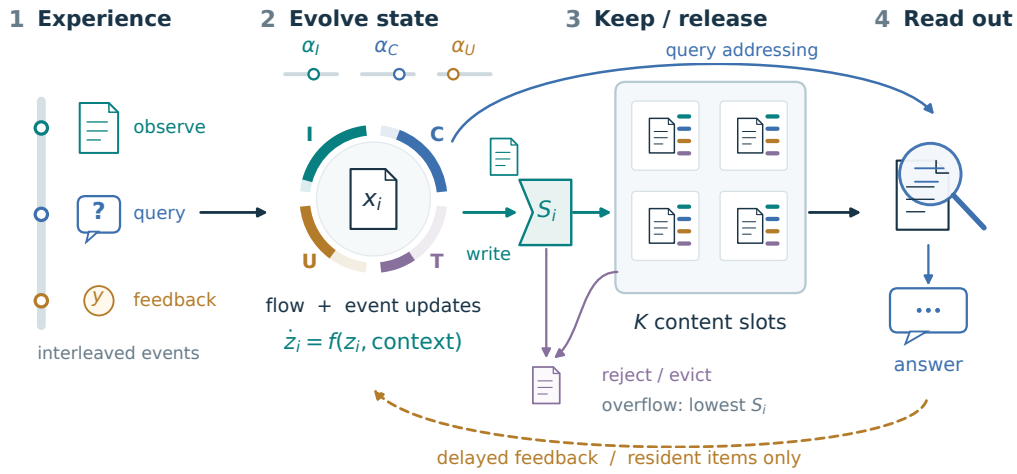


Figure 2: Prior held inputs drive flow before each event; new inputs are held forward. Only overflowing writes trigger removal. Retrieval reads resident content, and feedback arrives later.

Table 1: Constrained structured memory: eight slots, sixteen interference keys. Percentages except lag (target queries). All seven policies share the same streams; 24 worlds, three policy seeds. Full intervals accompany the machine-readable evidence (Appendix G).

Method	Acc.↑	Stale↓	Ret.↑	Lag↓
Fixed decay	46.48	0.00	46.64	12.88
Frequency	48.50	14.03	69.76	117.17
LRU	41.91	0.00	42.10	17.29
Frozen state	75.46	3.96	78.84	29.72
Replacement control†	54.93	1.81	74.84	103.77
Adaptive rule	71.02	0.01	79.09	42.33
AME	78.06	0.02	87.48	23.03

† Calibrated admission/lifetime control; distributional matching failed. This row is not an exactly matched counterfactual.

Methods share input tapes, initial records, slot limits, query budgets, and one overflow decision per eligible observation. Feedback arrives after the answer; the same corruption tape is used, but delivered correctness can differ because methods answer differently. Comparators are fixed exponential recency, retrieval frequency, LRU, frozen-state scoring, a calibrated replacement control, an adaptive rule, and AME. Frozen state retains learned insertion values but disables subsequent per-item changes. Fixed decay orders by most recent observation, whereas LRU also tracks retrieval. The replacement control adds a seeded admission gate; positive rescaling of decay alone cannot change the eviction order.

Every formal condition uses 24 schedule worlds and three policy/checkpoint seeds (11, 23, 42). Seeds are averaged within a world before inference. We report paired 95% percentile-bootstrap intervals (5,000 resamples) and sign-flip tests (10,000 draws), with Holm correction over each batch's declared comparisons. Repeated policy seeds are not independent worlds. C1–C4 comprise 17,568 policy trajectories; B1-A and B2–B7 add 5,184 and 7,560, respectively. Each batch has separate mini replay and frozen inputs. Protocol amendments and their preserved superseded evidence are disclosed in Appendix C.

Accuracy counts correct answers over all scored queries; stale error counts obsolete answers over the same denominator. Stable retention (SR) restricts correctness to stable queries. Adaptation lag is the first position at which two consecutive overlapping four-query windows are each at least 75% correct; unrecovered episodes are capped at their target-query horizon H. Adaptation score is $\text{AS} = 1 - \text{lag}/H$. It is a bounded first-passage summary, not proof of sustained recovery. Different episode lengths are never pooled without qualification.

5.2 RQ1: When does state evolution help?

With eight slots, AMEimproves overall accuracy and stable retention, but fixed decay adapts faster and already achieves zero stale error (Table 1). The capacity sweep changes both content slots and interference keys ($2K$), while keeping twelve core identities; it therefore measures *joint capacity/workload scaling*, not a pure scaling law. At four slots, AMEreaches 45.26% accuracy versus fixed decay's 20.23%. At sixteen slots, LRU reaches 98.81%, exceeding AME's 92.90%; at 64 slots, their accuracies are 98.99% and 98.86%. The advantage is strongest under constrained storage, not across every budget. Figure 7 shows all budgets. Fixed decay's zero stale error at eight slots coexists with poor stable retention; adaptive accuracy gains coexist with slower recovery. Evaluating the joint outcome vector prevents either observation from being mistaken for uniform improvement.

5.3 RQ2: Is replacement volume a sufficient explanation?

We calibrate an admission-gated recency policy on independent worlds using only eviction count and observed lifetime, never task outcomes. In the rapid-revision/high-interference attribution condition, its held-out means closely match AME:

Policy	Evictions	Mean lifetime	Acc. (%)	Lag
Fixed decay	184.10	25.42	44.60	16.58
AME	26.78	141.88	75.11	16.73
Replacement control	26.67	142.44	49.85	28.49

Against the calibrated control, accuracy rises by 25.26 points (Holm $p = 0.0132$) and lag decreases. Evictions are also fewer than under unthrottled recency. Replacement volume alone is insufficient to explain this comparison. Eight of nine cells pass the original 10% tolerance on the two means; one does not. Matching those means nevertheless leaves the identities and timing of removals uncontrolled. Observed lifetimes include right-censored residents and are partly coupled algebraically to replacement counts. Stronger checks of lifetime distributions in the subsequent validation matrix fail their matching gate. We therefore call those policies *calibrated replacement controls*, not exact causal counterfactuals. The result supports a restricted attribution beyond count alone; it does not eliminate replacement confounding.

5.4 RQ3: Can rate control navigate stability and plasticity?

We keep the head, event jumps, scoring formula, and stream fixed and set $\alpha_I = \alpha_C = \alpha_U = \alpha$, scanning $\mathcal{G} = \{0, 0.25, 0.5, 0.75, 1, 1.5, 2\}$. Conditions cross capacities 8/16 with mixed or rapid/high streams. Settings differ jointly in interference and revisions, not purely volatility. Testing all 21 rate pairs per condition on SR and AS yields 84 pairs and 168 jointly corrected endpoint tests. Twenty-seven pairs show opposite endpoint changes with both Holm-adjusted $p < 0.05$: nine in mixed demand at capacity 8, twelve in rapid/high at capacity 8, six in rapid/high at capacity 16, and none in mixed demand at capacity 16. These pairs share worlds and parameters; 27 is not a count of independent replications.

In rapid/high capacity 8, moving from $\alpha = 1$ to 2 raises AS from 0.472 to 0.669 ($\Delta = 0.197$, CI $[0.141, 0.251]$), while SR falls from 89.19% to 55.82% ($\Delta = -33.38$ points, CI $[-36.02, -30.67]$); both adjusted $p = 0.0168$. This is a direct operating-point trade-off without additional fitting. Some intermediate strengths are dominated, and retention saturates in mixed demand at capacity 16. Static comparators are fixed configurations, not equally extensive hyperparameter families: these experiments establish tunability of AME, not its uniqueness or superiority to every tunable heuristic.

5.5 RQ4: Which state components create the behavior?

State removal and flow removal answer different questions. We first pin one state to its neutral score value after every operation: $I = 1$, $C = 1$, $U = 0$, or $T = 0$. All weights remain fixed. Removing T removes its state and aging penalty, not elapsed-time integration of the remaining states. Usage removal produces the largest accuracy and retention loss, while yielding faster first-passage recovery. Consistency removal raises stale error from 0.07% to 8.76%. Importance removal increases retention but delays adaptation. These endpoint-specific effects are more informative than treating every component as an interchangeable accuracy improvement.

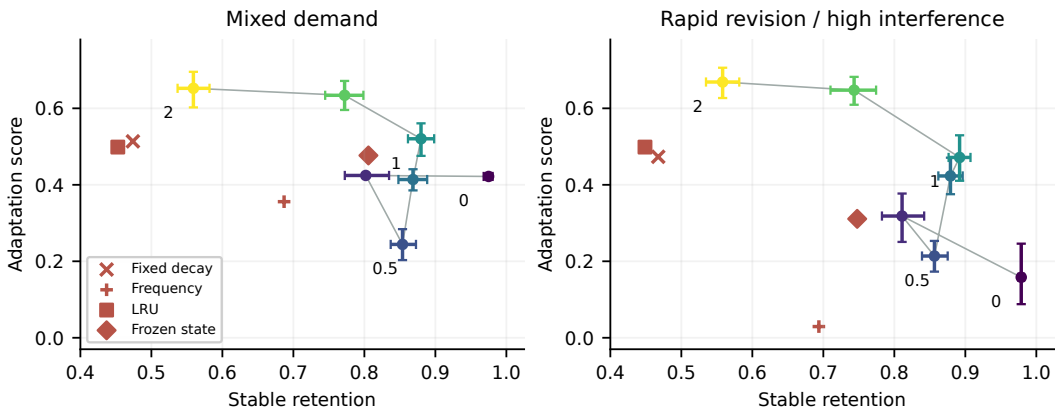


Figure 3: Eight-slot operating points: all seven rates, colored low (purple) to high (yellow), with selected labels and world-level 95% bootstrap intervals. Lines follow parameter order, not an optimized frontier. All conditions appear in Appendix F.

Table 2: State interventions under rapid revision and high interference (eight slots). No retraining; the removed state is pinned to its neutral score value. This experiment differs from the single-flow intervention in Figure 4.

Method	Acc.↑	Stale↓	Ret.↑	Lag↓
AME	75.02	0.07	88.25	15.53
No importance	67.14	0.02	91.70	26.72
No consistency	70.20	8.76	86.88	16.66
No usage	57.91	0.11	48.51	12.85
No temporal state	74.17	0.09	88.34	17.46

The second intervention enables only the importance, consistency, or usage *flow*, sweeping its strength over the same seven values. Other states still have their event updates, and T still ages. In rapid/high capacity 8 at strength 1, full evolution achieves SR 87.28%, AS 0.507, and stale error 0.022%. Importance-only evolution reaches 75.64%, 0.260, and 15.11%, respectively. Consistency-only and usage-only variants retain more stable knowledge but recover more slowly and return more stale answers; these nine endpoint contrasts pass the batch's Holm correction. Single-flow versus full removes two flows simultaneously, so this is evidence for joint behavior, not an orthogonal causal decomposition.

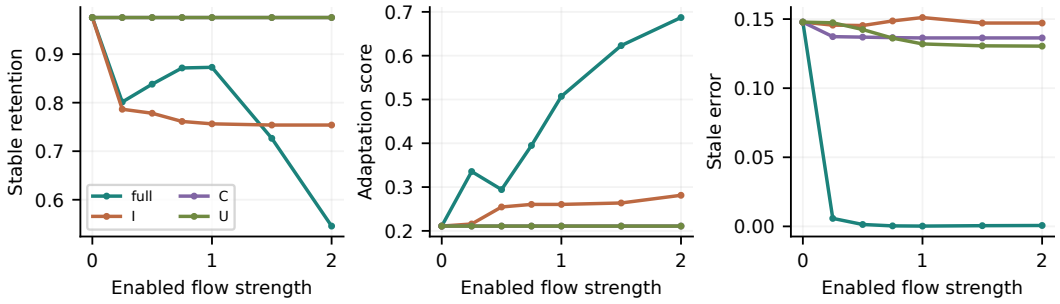


Figure 4: Single-flow interventions under rapid revision, high interference, and eight slots. "Full" enables all three flows; I/C/U enable only that flow. Discrete updates remain. All predefined strengths are shown, including unfavorable intermediate values.

Continuous integration is not the contribution. In a separate integration comparison, full evolution achieves 74.80% accuracy, stable-step Euler integration 74.18%, and event-only updates 67.26%. The corresponding stale errors are 0.09%, 0.09%, and 14.74%. These results distinguish

event-only behavior from the tested flow but do not establish a unique benefit of exact continuous integration. The event-only variant also removes inter-event T aging, so it is not identical to the zero-strength C1 policy. Adding a query-held context rule reaches 72.49%, below full evolution; the original controller already receives query-related features.

Resource cost. A serial CPU audit on the same 2,344-event schedule measures policy-call time and update counts, separately from trace serialization. State evolution entails $O(K)$ per-event resident work plus a small affine head. No speed or total-memory saving is claimed. Metadata, text, embeddings, and audit logs are outside a simple four-scalar accounting; Appendix E.1 reports the measured audit and its scope.

5.6 RQ5: Where does the mechanism fail?

Dormant knowledge is not the same as reacquisition. In a structured A–B–A cycle, pre-reminder recall is 0% for AMEand 68.75% for frequency. After new A observations, recovery lags are 12.82 and 10.76 queries. A high aggregate cycle accuracy (78.66% for AME) must not be interpreted as dormant preservation. Once content has been evicted, a later reminder can restore it through a new write; the state dynamics cannot reconstruct absent content.

Feedback and demand reliability. With 10%, 25%, and 50% correctness-feedback corruption, AMEaccuracy changes from a clean 75.33% to 74.75%, 72.10%, and 64.59%. Stable retention falls from 87.02% to 57.77% at 50% noise, while first-passage lag improves from 15.87 to 12.24 queries. Thus faster adaptation can accompany worse memory, even without an overall quality gain. Misleading demand cues reduce accuracy to 50.33%. These interventions flip feedback or topic cues; they do not model arbitrary natural-language observation noise.

Semantic addressing can defeat good retention. We replay confirmed public MemOps quotes in four controlled tasks using a frozen 384-dimensional sentence encoder [16, 18, 19]. Observations join a resident semantic key above cosine similarity 0.65; queries address the nearest resident cluster. This changes the addressing interface and demand affinity, not the state equations or weights. At eight slots, preference-update accuracy is 62.58% for AMEversus 48.76% for fixed decay, but stale preference error is 18.87% versus 0.03%. At sixteen slots, accuracy reverses to 73.39% versus 88.57%. Correct content is resident for 99.25% of AMEqueries, yet 25.04% of all queries retrieve an old value despite that availability. A post-hoc trace audit links only 19.79% of 288 eligible revisions to the previous resident semantic key. Version splitting prevents consistency dynamics from recognizing the relation. This diagnostic was not used to tune thresholds or rerun favorable policies.

Real conversational transfer remains weak. A fixed 24-question subset per dataset, three seeds, and one Agnes backend yield judged correctness 4.17% for AMEversus 15.28% for fixed decay on LongMemEval, and 11.11% versus 3.33% on LoCoMo. Intervals are wide. MemoryBench-NFCats gives normalized rubric quality 0.740 versus 0.733, not comparable to binary QA accuracy. These are custom subset results, not official benchmark scores. The reader and judge share the same model, backend variation is untested, and absent version labels prevent reliable stale-error measurement in these datasets.

Table 3: Capability boundaries supported by the tested settings. These are conditional observations, not guarantees for all environments with the same informal label.

Condition	Observed boundary
Small capacity / interference	Accuracy and stable-retention gains; adaptation need not improve.
Larger content budget	Simple recency can match or outperform AME.
Rate intervention	Opposing endpoint changes in some settings; nonmonotonic points remain.
Dormant knowledge return	Evicted records require new evidence; frequency may preserve more.
Unreliable signals	Corruption and misleading demand degrade quality.
Semantic versions	Correct content availability is insufficient when keys split.
Long revised streams	Reduced usage inertia; no general infinite-horizon guarantee.

6 Discussion

What makes the mechanism useful? Past utility and present validity are different signals. Frequency can protect heavily used but obsolete records; recency can discard stable facts while keeping recent interference. The state score represents both signals and lets them change on different time scales. In particular, usage records evidence of prior usefulness while its decay limits persistence of that evidence, and consistency can discount a superseded version without assuming that every old fact is obsolete. The component interventions support this interpretation, not a uniquely correct or minimal decomposition.

The rate experiment supplies the strongest evidence because content, weights, and scoring are unchanged within each world. Changing a rate changes the state history that reaches a budget decision. The intervention acts through admission and eviction, which determine available information. The attribution question is whether aggregate replacement volume alone explains performance; the present evidence rejects that explanation in specific mean-matched comparisons, while leaving finer replacement counterfactuals open.

What survives over longer horizons? The long-horizon test uses exact 500/1,000/5,000-event prefixes with a context switch every 97 events and renewed observed fact revisions after each switch. At 5,000 events and eight slots, full evolution reaches 75.21% accuracy versus 43.32% for fixed decay ($\Delta = 31.89$ points, CI $[31.46, 32.33]$, Holm $p = 0.0090$). Stable retention is 98.43% versus 45.51%; stale error is zero for both. At sixteen slots, accuracy is 87.19% versus 79.37%, but adaptation is slower by 0.387 target queries (CI $[0.320, 0.461]$, adjusted $p = 0.0090$). Thus even this favorable long-stream comparison retains a cost.

Frequency's stale-answer rate at sixteen slots increases from 19.09% at 500 events to 42.39% at 5,000; full evolution is 0.0042% at 5,000. The largest observed usage state is 220 without I/C/U flow and approximately 5.29 at strength 1. These are finite-stream measurements, consistent with the contraction of the usage flow. They neither prove infinite-horizon stability nor demonstrate bounded open-world metadata: the stress environment has only 28 fact identities. Full time curves, the collapse diagnostic, and selected-item trajectories are reported in Appendix E.

Stability–plasticity as a policy choice. Competing goals here do not establish a universal impossibility theorem: some interventions improve multiple metrics. Practical control would require choosing an acceptable retention loss or stale-error constraint before selecting an operating point, ideally using independent validation. Our grid neither selects deployment strengths automatically nor establishes global Pareto optimality. An equally budgeted sweep of recency–frequency mixtures is an important untested comparator family.

Scope and deployment limitations. Structured identities, observed revision signals, and explicit post-answer feedback simplify the synthetic problem. Shared public templates limit independence of content, although schedules are held out. Semantic migration is a controlled replay, not an autonomous tool-using agent. The conversational results use one backend and one same-model judge, too few questions for strong transfer claims, and no competitive full-system study against current agent-memory methods. Storage is constrained in records rather than total bytes; state updates add work. These limitations constrain the paper's claim to a measured mechanism and capability boundary. Deployment additionally needs conflict detection, deletion semantics, provenance, and privacy controls.

7 Conclusion

Adaptive Memory Evolution makes the retention state of individual content records an explicit dynamical object. With frozen controller weights, rate interventions alter stable retention and post-shift adaptation, providing statistically supported operating-point choices in constrained non-stationary streams. Component and long-horizon analyses connect these choices to usage inertia and observed consistency, while dormant-recall and semantic-transfer failures expose their limits. The evidence supports a tunable memory-management mechanism; it does not establish LNN superiority, continuous integration necessity, universal improvement, or a general solution to lifelong agents.

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A State Properties and Exact Implementation

Proposition 1 (Invariant state ranges under the implemented updates)**.** *Suppose* $k > 0$*,* $\alpha_I, \alpha_C, \alpha_U \geq 0$*,* $q \in [0,1]$*,* $w \in (0,2)$*, and delivered feedback* $y \in [0,1]$*. Starting from the prescribed bootstrap or insertion state, the hybrid updates preserve* $0 < I \leq 20$*,* $0 \leq C \leq 1$*,* $U \geq 0$*, and* $T \geq 0$ *at finite event times. Under held inputs and positive corresponding strengths,* I *and* C *approach their targets monotonically and* U *contracts.*

Proof. Each update of I or C in Equation 3 is a convex combination of its pre-flow value and target because its exponential coefficient lies in $[0,1]$. The importance target lies in $[0.5, 2.5)$ and the consistency target in $\{0,1\}$. Insertion has $I = gw \in (0,2)$; duplicate observation caps its positive increment at 20. Consistency observation resets to one and feedback is a convex combination with y. Usage flow multiplies a nonnegative value by a nonnegative factor and feedback adds a nonnegative value. Temporal flow adds a nonnegative interval and resets to zero. These operations preserve the asserted ranges by induction over events. The differences from held targets are multiplied by positive decaying exponentials, establishing monotone approach between events. This does not imply monotone task performance or bounded T across an infinite stream. □

Proposition 2 (Conditional usage bound)**.** *If* $\alpha_U k(t) \geq \kappa > 0$ *while a record is resident, usage increments are at most one, and successive increment times are separated by at least* $\delta > 0$*, then post-increment usage satisfies*

$$U_n \leq r^n U_0 + \frac{1 - r^n}{1 - r}, \qquad r = e^{-\kappa\delta/20} < 1. \tag{6}$$

Proof. Between successive increments, the flow contracts usage by at most r; hence $U_{n+1} \leq rU_n + 1$. Iterating this scalar inequality gives the bound. Additional resets through eviction and reinsertion only lower usage. If strength is zero or increments can accumulate without a positive spacing bound, the uniform bound does not follow. The observed usage of 220 at zero strength is therefore not evidence of numerical instability. □

Rate control and selection discontinuities. The state at a fixed time is smooth in a rate on any interval without discrete policy changes. The map from a rate to task outcomes is not smooth in general: when two retention scores cross, a different record can be evicted and all later observations, answers, and feedback-dependent states can diverge. Consequently, neither the smooth vector field nor the finite C1 grid implies a continuous monotone performance frontier. Our operational claim is supported opposing changes at sampled settings.

Why decay rescaling cannot match replacement by itself. For fixed observation ages a_i, every positive λ gives the same ordering of $\exp(-\lambda a_i)$, equivalently $-\lambda a_i$. A fixed-capacity minimum-score eviction policy therefore chooses the same victim under any positive rescaling, except unchanged ties. The implemented replacement control explicitly changes admission, not merely the decay coefficient. The matrix extension also uses a minimum resident age. This distinction is necessary for interpreting the control's name.

Lifetime bookkeeping. If each admitted residency is followed to eviction or the evaluation endpoint, summing its observed duration gives $\int_0^T |\mathcal{M}_t|\, dt$. When the store is always full, this is KT. With E evictions and K final right-censored residents, mean observed residency duration is $KT/(E+K)$. Thus matching mean duration and eviction count does not provide two independent controls under full occupancy. Warm-up occupancy, the lifetime distribution, censoring, resident age, and the identities of removed items remain relevant.

B Frozen Head and Feature Provenance

The head is an affine map from 40 features to three coefficients, not an LNN or recurrent hidden-state model. The first 32 features encode content. The remaining features are: an observation/context event indicator; $\log(1+\Delta)/5$; exponentially decayed query demand for the current key; capped time since its last query; an exponential mean of observed version conflict; mean delivered feedback; most

recent delivered feedback; and agreement with the held context. Demand decays with time constant 30. Conflict and feedback running averages are causal. No stable-role flag or reference answer is included.

The earlier development run used four conditions (mixture, feedback noise, misleading feedback, and cyclic demand), with 8 training and 4 validation schedule seeds per condition and capacities 8/12. Noise-related conditions share underlying worlds. The training objective was

$$\mathcal{L} = (1 - A) + 0.25(1 - \mathrm{SR}) + 0.25\mathrm{SE}. \tag{7}$$

Hard-capacity decisions were trained with REINFORCE, a deterministic fixed-policy baseline, AdamW at 0.003, batches of eight, gradient clipping at one, and sixteen epochs. Validation utility pooled the development conditions and capacities. Initialization was eligible for selection. The selected checkpoints for seeds 11/23/42 are epochs 15/9/14, with validation utilities 0.7467/0.7412/0.7657. These weights precede the validation matrix and were not retrained for any reported B/C batch. This provenance does not make the synthetic content distribution independent of development templates.

In the rule baseline, $(g, w, k) = (2/3, 1.5, 0.3)$. For learned evolution, g is the insertion gate; it must not be confused with the externally swept α. We use a shared strength in C1, single active flow in C2, default strength one in semantic migration, and strengths $\{0, 0.25, 1, 2\}$ in long-horizon stress. All content versions are explicit records. The version ledger persists identifiers after eviction but not content; the query history is also auxiliary metadata outside the slot constraint.

C Protocols, Statistics, and Amendments

Frozen batch design. Protocols, checkpoints, random seeds, event streams, full state trajectories, write rejection decisions, censored lifetimes, raw metrics, and statistical comparisons are stored with hashes. The final characterization used 244 mini configurations, each exactly replayed, before 17,568 formal policy trajectories. The current source test suite passes 251 tests. These checks establish implementation and replay consistency, not external validity or absence of every possible modeling bias. Protocols were frozen locally before each formal batch; this was an iterative research project, not an externally registered study blind to all earlier results.

Experiment family	Formal worlds	Seeds	Scope
B1-A attribution	72001–72024	11/23/42	3 dynamics $\times$3 pressure cells
B2 capacity	82100–82123	11/23/42	$K = 4, 8, 16, 32, 64$
B3 cycle	83100–83123	11/23/42	A–B–A with pre-reminder probes
B4 corruption	84100–84123	11/23/42	clean, 10/25/50%, misleading cue
B5 revision	85100–85123	11/23/42	observed value updates
B6 state intervention	86100–86123	11/23/42	four neutral-state controls
B7 integration	87100–87123	11/23/42	event-only, Euler, query-held cue
C1 rate scan	91100–91123	11/23/42	two settings, two capacities
C2 flow attribution	92100–92123	11/23/42	four flow families, seven strengths
C3 semantic replay	93100–93123	11/23/42	four tasks, two capacities
C4 long horizon	94100–94123	11/23/42	three lengths, two capacities

The structured matrix uses shared four-record bootstrap states, eight currently active facts, and interleaved observation/query/feedback. Stable facts are periodically queried; two groups of four volatile facts exchange demand. B1-A varies interference pools 2/8/16 and stable/slow/rapid revision schedules. B2 fixes the core identities but scales interference pool size to $2K$, with equal core and interference arrival counts per phase. B3 excludes factual revisions to isolate demand return. B4 shares the same observation and query tapes across corruption rates. C1 mixed and rapid/high use different interference pool sizes and revision settings; an effect across them cannot be attributed to one factor alone.

What is held equal? Every policy sees the same exogenous observations, context cues, query times, and corruption draws within a world. Scored feedback is generated only after answering, so identical exogenous streams do not imply identical realized feedback histories. The core synthetic retrieval budget is one answer from the addressed resident key. A new observation creates one capacity decision; a duplicate refresh does not create an extra resident version. Policies may differ

in accepted writes and rejections, which are recorded separately from resident replacements. The common implementation includes four state fields even in fixed policies, so multiplying four scalars by capacity is not a measured incremental heap allocation.

Metrics and inference. For query set Q and stable subset Q_s, $A = |Q|^{-1} \sum_q \mathbf{1}[\hat{a}_q = a_q]$, $\mathrm{SR} = |Q_s|^{-1} \sum_{q \in Q_s} \mathbf{1}[\hat{a}_q = a_q]$, and $\mathrm{SE} = |Q|^{-1} \sum_q \mathbf{1}[\hat{a}_q$ is an obsolete value of the queried fact]. Missing answers and wrong-fact answers are incorrect but not automatically stale. The lag rule needs at least five queries; an episode with no recovery takes its full horizon. A recovered final query and an unrecovered episode may share the same lag, so recovery success is separately saved. Intervals and tests use paired worlds after averaging the three policy seeds; neither individual queries nor seeds inflate the sample size. C1 corrects 168 endpoint tests, C2 252, C3 80, and C4 90 within their respective batches. These families were not jointly corrected into one universal claim. The archive retains both within-batch and additional across-matrix comparisons where available.

Amendments without policy optimization. The semantic replay initially counted unchanged continuation phases as adaptation episodes. Evaluation was corrected to start after observed new-value exposure and to distinguish demand return; the same seeds were rerun, retaining the earlier files. Twelve deterministically selected old/new trajectory replays have identical actions, states, queries, and lifetimes; the changed fields are adaptation episode definitions and derived metrics. The long-horizon prototype repeatedly changed demand but concentrated fact revisions early. Before interpreting formal performance, it was amended to introduce a different observed public value on the first active-fact observation after every subsequent context shift. Prior mini and partial formal files remain in the superseded archive. These are protocol corrections, not evidence of a fully preregistered end-to-end study.

D Semantic and Language Transfer Details

Controlled public-text migration. MemOps (revision `312af65e2c7b`, full hash in the archived protocol) supplies confirmed observation/update quotes. Ten target domains cover browser, keyboard, author, hotel brand, naming preferences, travel, examination date, and storage plan. Four stable controls are sampled from never-revised side facts. Each task has four active targets and sixteen interference identities. Public quotes are reused across 24 schedule worlds; uncertainty thus describes schedule variation, not sampling of independent language corpora. Historical distractors explicitly say they are no longer current, but the adapter has no gold semantic contradiction resolver.

The encoder is frozen `all-MiniLM-L6-v2`. A new observation hashes its content for duplicate detection. Otherwise its nearest resident embedding is used to reuse a key if cosine similarity is at least 0.65; failing that, a new key is assigned. Reused keys increment the observed arrival version. A query selects the nearest resident embedding's key, then the newest resident version within that key. All thresholds were fixed before formal outcomes. Demand affinity is $\max(0, \cos(x_i, c))$ under the held context cue, replacing the structured binary affinity. The 384-dimensional vector is projected to the same 32 head features. Removed records lose their embeddings; no evicted vector archive can supply answers. The evaluation wrapper's gold identity, value, and stability label are excluded from policy inputs.

Four tasks test preference update, contradictory historical evidence, old-preference revival, and temporary task information. Preference-update and contradiction adaptation include only the observed revised-value queries in the actual update phase; unchanged continuation is not counted as another update. Revival includes a probe before new reminder observations. At capacity sixteen, pre-reminder recall is 17.71% for AMEversus 98.61% for frequency, despite AME's higher aggregate revival accuracy. Mismerge counts track wrong-fact key merging, not false separation of versions. The post-hoc semantic routing audit reports both; its descriptive link rate is conditional on an old value still being resident. It is not a new randomized intervention on the addressing layer.

Conversational transfer protocol. The held-out subset comprises 24 LongMemEval questions (four from each of six categories), 24 LoCoMo questions drawn across ten conversations, and 24 NFCats test questions with 200 released training dialogs. Each is evaluated with seven policies and three policy seeds, for 1,512 formal answer records. LoCoMo answers are clustered by conversa-

tion; other datasets by question. All NFCats questions share one training history, so uncertainty is conditional on that history. A held-out question is not presented during consolidation.

The memory adapter uses exhaustive spans up to 1,200 characters, capacity 64, top-five retrieval, and a 12,000-character reader budget. The topic signal is a fixed content-projection heuristic, not gold semantic parsing. LongMemEval/LoCoMo provide no correctness feedback during consolidation; NFCats supplies released off-policy dialog feedback. Dated dialogs use days as the time unit, while NFCats uses dialog steps; this time-scale transfer is uncalibrated. These limitations make the transfer study a boundary probe rather than a production memory system comparison.

Answers and blinded evaluation use `agnes-3.0-flash`, temperature zero, answer maximum 384 tokens, at https://api.agnes-ai.cn/v1. Rate-limited calls, retries, prompts, raw responses, token usage, and judge-format repairs are saved. LongMemEval/LoCoMo use reference-based binary correctness plus auxiliary token F1. NFCats uses a reference-free 1–5 rubric normalized as $(s-1)/4$; satisfactory response rate uses $s \geq 4$. Neither NFCats quality nor its uncertainty should be interpreted as binary QA accuracy. We did not vary the language backend or use a separate judge model. Stale error, stable retention, and adaptation lag are unavailable without suitable labels and are left null, not inferred from generic wrong answers.

E Long-Horizon and Computational Diagnostics

The stress generator maintains exact common prefixes at 500, 1,000, and 5,000 events. Following 37 warm-up events, each phase has one context cue and 32 observation/query/feedback triples, hence 97 events. Every complete post-initial phase introduces four active-fact revisions upon their first observation. Each shift has sixteen target queries. The terminal partial episode is excluded from the lag average but retained in raw event metrics. No missing future observation is imputed.

The descriptive collapse indicator is two consecutive 100-event scored windows with accuracy below 10%. At 5,000 events the fully frozen-state control triggers it in 16/72 trajectories at capacity eight and 2/72 at capacity sixteen; no other tested method does. This indicator does not certify quality: frequency still has substantial stale-answer accumulation. At sixteen slots, its final-window obsolete-resident fraction is 74.31%, versus 0.039% for strength one. All state values remain finite. These results concern the finite 28-key test and do not establish infinite-horizon or open-world guarantees.

E.1 Resource accounting

The serial computational audit replays fixed environments separately from the parallel formal experiment. Policy time includes calls for state update, observation, retrieval, eviction, and feedback; it excludes trace serialization and language-model inference. Counts and Python object-graph estimates are saved in `serial_computational_audit.json`. These estimates include object bookkeeping and, in the full policy graph, retained lifetime logs; they are not isolated deployment RSS. A theoretical four-float64 state payload is 32 bytes per record, but common baseline records also allocate these fields. We therefore do not claim a measured 32-byte incremental cost or an efficiency advantage.

For 5,000-event stress trajectories at eight slots, strength one averages 39,959 item-state updates and 5,000 coefficient evaluations. Observed mean policy time is approximately 0.099 seconds versus 0.041 seconds for fixed decay, measured within four concurrent CPU workers. At sixteen slots these are approximately 0.131 and 0.050 seconds. They indicate extra work but are not clean speed benchmarks. Semantic addressing further requires resident embeddings and an encoder; offline embedding generation is excluded from policy-call time. All versions of the experiment retain growing auxiliary metadata or audit logs, so fixed content capacity must not be described as a strict total-memory guarantee.

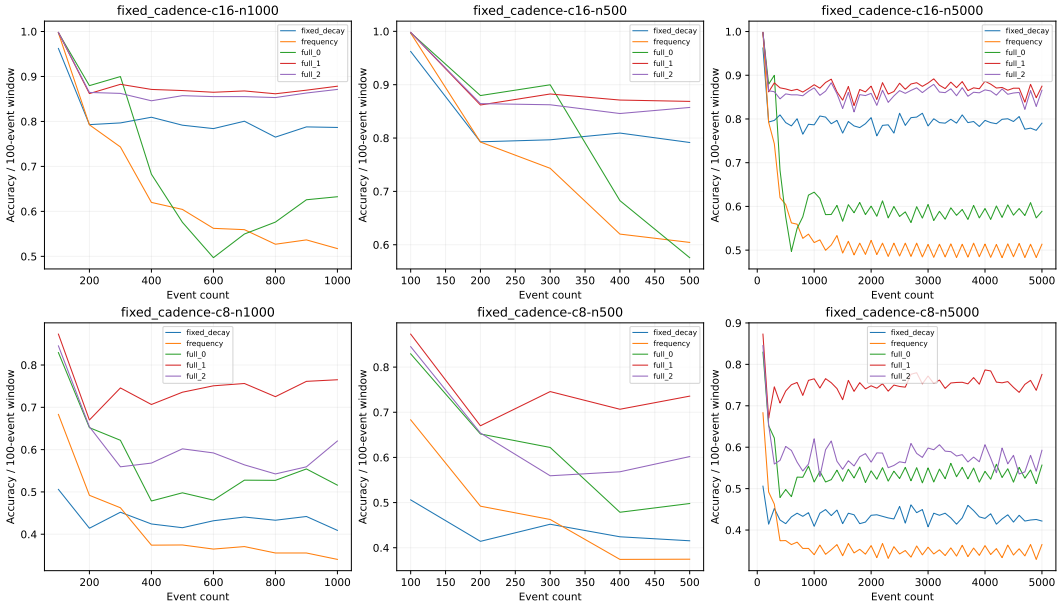


Figure 5: All three predefined horizons and two capacities. Curves average 100-event window accuracy across frozen worlds and seeds. Shorter runs are exact prefixes; different panels must not be treated as independently generated environments.

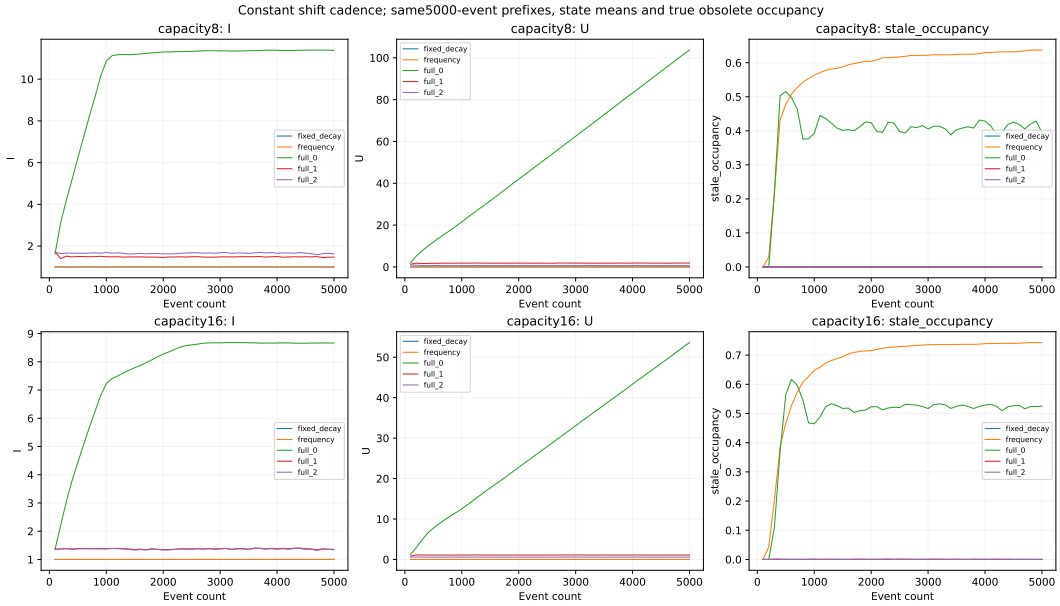


Figure 6: Long-horizon mean importance, usage, and obsolete-resident occupancy. Zero-strength usage growth follows the absence of usage decay. Fixed-policy state values are common implementation placeholders, not learned representations.

F Complete Operating Points and State Traces

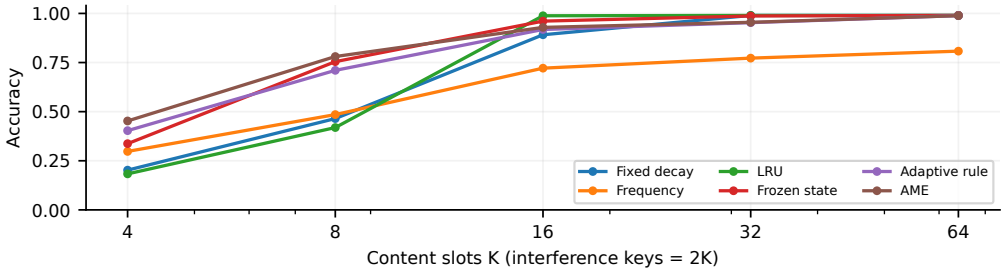


Figure 7: Capacity and interference jointly scale. All predefined budgets are shown; the larger-budget reversal rules out universal superiority. Complete replacement-control rows and intervals accompany the evidence files.

The main plot emphasizes eight-slot conditions for legibility; the following figure also includes both sixteen-slot conditions. The empirical nondominated envelope is computed only over this finite comparison set. Bootstrap frontier membership in the archive measures resampling stability, not a posterior probability of global optimality. All seven rates, including dominated points, remain in the tables.

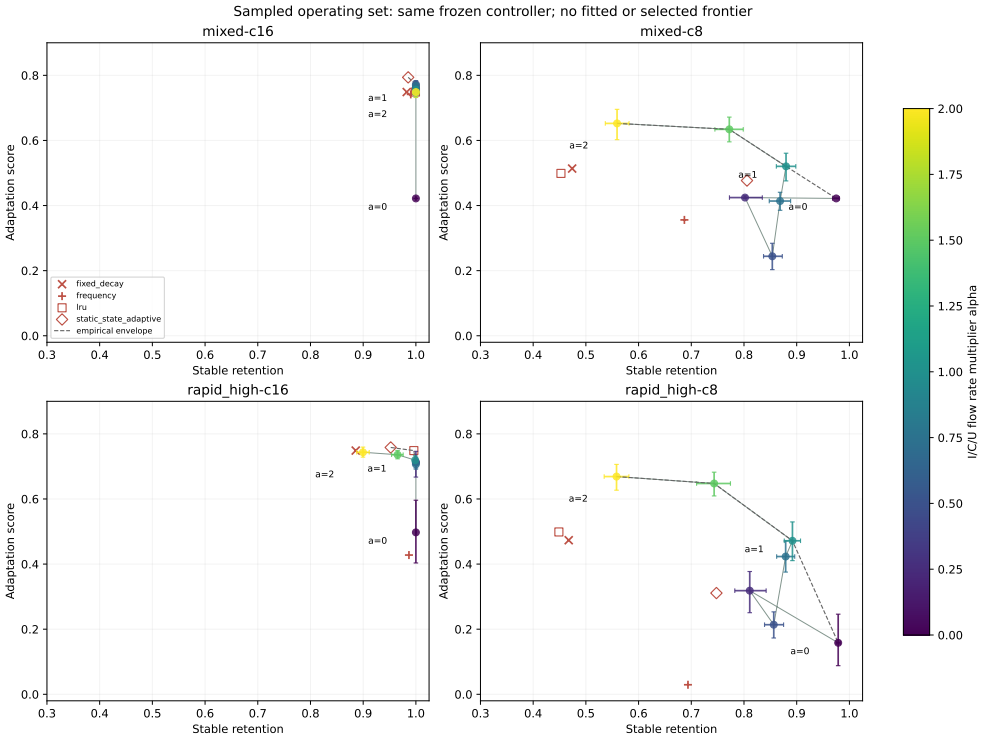


Figure 8: All four C1 conditions. Capacity sixteen mixed demand saturates stable retention and has no rate pair with statistically supported opposing endpoint changes.

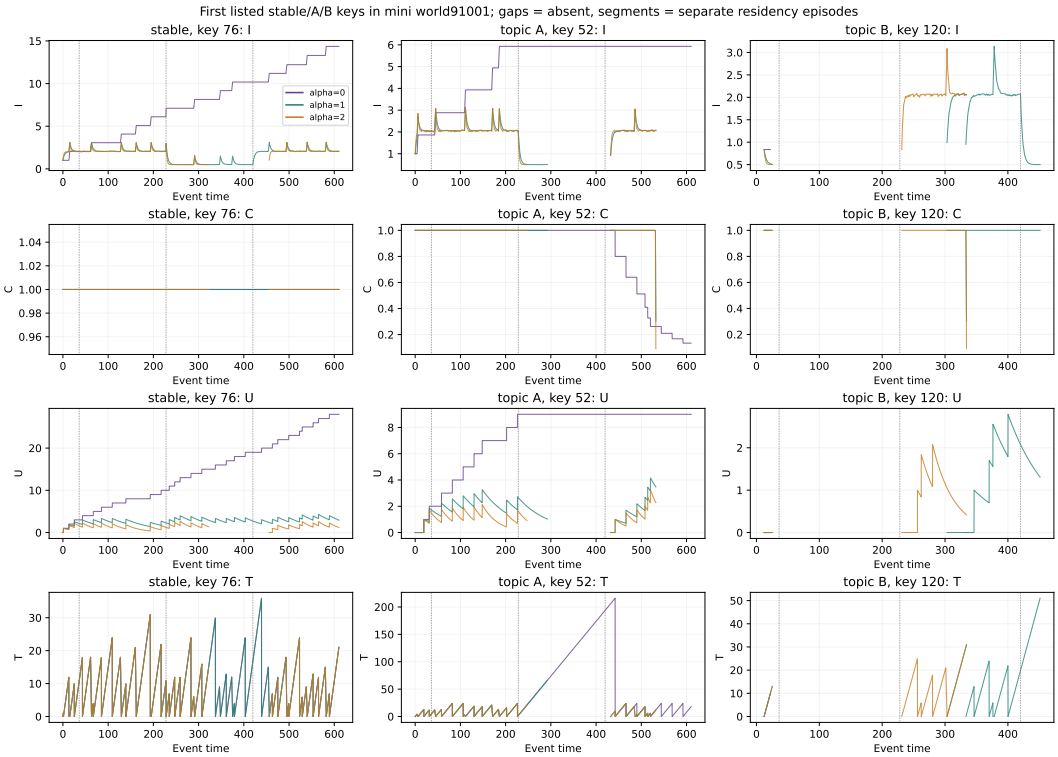


Figure 9: Predefined first stable/A/B keys in mini world 91001 at strengths 0/1/2. All four states are shown. Missing segments denote absence from the resident store; separate residency episodes are not joined. Reappearance can therefore be distinguished from preservation of dormant content.

G Complete Numerical Results

All rates and methods are retained. Main-text percentages are rounded from the frozen JSON summaries, not copied by hand into table cells. The evidence directory contains full world-level 95% intervals and paired comparisons; the following compact tables show means for every condition. Their experimental batch identifiers are included only to connect each row to a frozen protocol, not as the main scientific organization of the paper. B1-A's full nine-cell attribution summary is also included in the evidence directory. Replacement-control rows do not imply successful lifetime-distribution matching.

Table 4: B2: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
capacity4	Fixed decay	20.23	0.00	19.89	87.15
capacity4	Frequency	29.76	6.78	47.49	128.00
capacity4	LRU	18.33	0.00	18.03	101.96
capacity4	Frozen state	33.70	3.92	36.41	107.77
capacity4	Replacement control†	26.88	0.61	34.78	122.90
capacity4	Adaptive rule	40.33	0.01	46.30	101.67
capacity4	AME	45.26	0.00	55.47	96.32
capacity8	Fixed decay	46.48	0.00	46.64	12.88
capacity8	Frequency	48.50	14.03	69.76	117.17
capacity8	LRU	41.91	0.00	42.10	17.29
capacity8	Frozen state	75.46	3.96	78.84	29.72
capacity8	Replacement control†	54.93	1.81	74.84	103.77
capacity8	Adaptive rule	71.02	0.01	79.09	42.33
capacity8	AME	78.06	0.02	87.48	23.03
capacity16	Fixed decay	89.14	0.00	89.75	8.38
capacity16	Frequency	72.14	18.36	100.00	84.42
capacity16	LRU	98.81	0.00	99.74	8.23
capacity16	Frozen state	96.08	1.40	97.65	9.60
capacity16	Replacement control†	77.74	4.35	99.70	71.71
capacity16	Adaptive rule	91.82	0.07	97.57	8.52
capacity16	AME	92.90	0.07	99.35	8.81
capacity32	Fixed decay	98.97	0.00	99.99	8.23
capacity32	Frequency	77.25	17.16	100.00	55.28
capacity32	LRU	98.99	0.00	100.00	8.23
capacity32	Frozen state	98.71	0.23	99.48	8.09
capacity32	Replacement control†	91.31	3.77	100.00	21.81
capacity32	Adaptive rule	95.30	0.03	99.41	8.52
capacity32	AME	95.48	0.03	100.00	8.69
capacity64	Fixed decay	98.99	0.00	100.00	8.23
capacity64	Frequency	80.80	15.42	100.00	38.20
capacity64	LRU	98.99	0.00	100.00	8.23
capacity64	Frozen state	99.00	0.14	99.63	7.42
capacity64	Replacement control†	97.36	1.00	100.00	10.03
capacity64	Adaptive rule	98.88	0.00	99.95	8.50
capacity64	AME	98.86	0.00	100.00	8.69

Table 5: B3: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
cyclic	Fixed decay	44.18	0.00	46.09	18.29
cyclic	Frequency	60.39	0.00	70.04	22.19
cyclic	LRU	42.03	0.00	43.14	19.21
cyclic	Frozen state	74.74	0.00	80.43	16.58
cyclic	Replacement control†	56.29	0.00	68.97	28.59
cyclic	Adaptive rule	70.81	0.00	80.77	23.46
cyclic	AME	78.66	0.00	86.76	16.65

Table 6: B4: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
clean	Fixed decay	45.31	0.00	45.62	15.69
clean	Frequency	49.01	11.96	68.48	29.45
clean	LRU	43.14	0.00	43.84	17.33
clean	Frozen state	65.91	3.60	75.00	22.88
clean	Replacement control†	51.74	2.01	67.26	28.10
clean	Adaptive rule	71.42	0.02	83.33	19.60
clean	AME	75.33	0.05	87.02	15.87
noise10	Fixed decay	45.31	0.00	45.62	15.69
noise10	Frequency	49.01	11.96	68.48	29.45
noise10	LRU	43.14	0.00	43.84	17.33
noise10	Frozen state	65.75	3.67	75.20	23.06
noise10	Replacement control†	51.74	2.01	67.26	28.10
noise10	Adaptive rule	69.51	0.15	77.04	17.88
noise10	AME	74.75	0.05	83.81	15.24
noise25	Fixed decay	45.31	0.00	45.62	15.69
noise25	Frequency	49.01	11.96	68.48	29.45
noise25	LRU	43.14	0.00	43.84	17.33
noise25	Frozen state	65.34	3.79	73.90	22.31
noise25	Replacement control†	51.74	2.01	67.26	28.10
noise25	Adaptive rule	65.21	0.07	65.71	14.27
noise25	AME	72.10	0.07	76.32	13.60
noise50	Fixed decay	45.31	0.00	45.62	15.69
noise50	Frequency	49.01	11.96	68.48	29.45
noise50	LRU	43.14	0.00	43.84	17.33
noise50	Frozen state	65.08	3.83	73.84	22.69
noise50	Replacement control†	51.74	2.01	67.26	28.10
noise50	Adaptive rule	60.66	0.07	50.65	12.04
noise50	AME	64.59	0.09	57.77	12.24
misleading_importance	Fixed decay	45.31	0.00	45.62	15.69
misleading_importance	Frequency	49.01	11.96	68.48	29.45
misleading_importance	LRU	43.14	0.00	43.84	17.33
misleading_importance	Frozen state	67.49	5.24	85.37	26.47
misleading_importance	Replacement control†	51.74	2.01	67.26	28.10
misleading_importance	Adaptive rule	48.03	0.00	78.82	29.79
misleading_importance	AME	50.33	0.01	80.92	30.17

Table 7: B5: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
version_update	Fixed decay	45.05	0.00	45.92	17.92
version_update	Frequency	50.65	11.31	72.80	29.82
version_update	LRU	43.42	0.00	44.44	17.04
version_update	Frozen state	66.59	3.41	73.94	20.92
version_update	Replacement control†	51.40	2.39	66.62	26.96
version_update	Adaptive rule	69.31	0.09	82.42	20.98
version_update	AME	74.42	0.09	87.86	16.31

Table 8: B6: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
rapid_high	Fixed decay	44.64	0.00	45.36	17.29
rapid_high	Frequency	50.46	11.48	70.69	29.42
rapid_high	LRU	42.69	0.00	43.40	19.19
rapid_high	Frozen state	66.89	3.05	73.80	20.74
rapid_high	Replacement control†	51.95	1.81	66.71	27.51

Condition	Policy	Acc.	Stale	Ret.	Lag
rapid_high	Adaptive rule	68.51	0.09	79.04	19.62
rapid_high	AME	75.02	0.07	88.25	15.53
rapid_high	No importance	67.14	0.02	91.70	26.72
rapid_high	No consistency	70.20	8.76	86.88	16.66
rapid_high	No usage	57.91	0.11	48.51	12.85
rapid_high	No temporal state	74.17	0.09	88.34	17.46

Table 9: B7: complete structured results (percentages; lag in queries).

Condition	Policy	Acc.	Stale	Ret.	Lag
rapid_high	Fixed decay	44.94	0.00	46.44	17.06
rapid_high	Frequency	49.83	11.48	69.59	28.60
rapid_high	LRU	43.08	0.00	44.70	18.71
rapid_high	Frozen state	66.44	3.88	75.33	20.81
rapid_high	Replacement control†	50.92	1.98	65.81	28.31
rapid_high	Adaptive rule	69.18	0.09	83.33	21.48
rapid_high	AME	74.80	0.09	87.27	16.01
rapid_high	event_only	67.26	14.74	98.31	26.38
rapid_high	discrete_euler	74.18	0.09	86.92	17.05
rapid_high	query_conditioned	72.49	0.08	85.14	19.88

Table 10: Fixed-subset language transfer. Quality is binary judged correctness for conversational QA and normalized rubric quality for NFCats; these scales must not be pooled.

Dataset	Policy	Quality	95% CI	Evidence hit
LoCoMo	Fixed decay	0.033	[0.000,0.100]	0.067
LoCoMo	Frequency	0.100	[0.017,0.233]	0.094
LoCoMo	LRU	0.050	[0.000,0.133]	0.067
LoCoMo	Frozen state	0.011	[0.000,0.033]	0.061
LoCoMo	Replacement control†	0.044	[0.000,0.122]	0.089
LoCoMo	Adaptive rule	0.094	[0.000,0.211]	0.094
LoCoMo	AME	0.111	[0.017,0.222]	0.111
LongMemEval	Fixed decay	0.153	[0.028,0.306]	0.167
LongMemEval	Frequency	0.069	[0.028,0.125]	0.736
LongMemEval	LRU	0.139	[0.014,0.278]	0.208
LongMemEval	Frozen state	0.111	[0.028,0.208]	0.528
LongMemEval	Replacement control†	0.250	[0.097,0.417]	0.278
LongMemEval	Adaptive rule	0.083	[0.000,0.208]	0.264
LongMemEval	AME	0.042	[0.000,0.111]	0.306
NFCats	Fixed decay	0.733	[0.583,0.854]	–
NFCats	Frequency	0.736	[0.618,0.840]	–
NFCats	LRU	0.740	[0.604,0.865]	–
NFCats	Frozen state	0.642	[0.507,0.767]	–
NFCats	Replacement control†	0.743	[0.611,0.858]	–
NFCats	Adaptive rule	0.660	[0.517,0.788]	–
NFCats	AME	0.740	[0.632,0.840]	–

Table 11: C1: all frozen operating points. Percentages except adaptation score.

Condition	Policy	Acc.	Stale	Ret.	Adapt.
mixed-c16	fixed_decay	95.29	0.00	98.26	0.749
mixed-c16	frequency	93.99	0.00	99.03	0.742
mixed-c16	lru	96.16	0.00	99.91	0.749
mixed-c16	static_state_adaptive	96.70	0.00	98.54	0.794

Condition	Policy	Acc.	Stale	Ret.	Adapt.
mixed-c16	full_0	83.33	0.00	100.00	0.422
mixed-c16	full_0.25	97.09	0.00	100.00	0.763
mixed-c16	full_0.5	96.73	0.00	100.00	0.772
mixed-c16	full_0.75	96.65	0.00	100.00	0.768
mixed-c16	full_1	96.44	0.00	100.00	0.756
mixed-c16	full_1.5	96.15	0.00	100.00	0.747
mixed-c16	full_2	96.20	0.00	100.00	0.749
mixed-c8	fixed_decay	45.96	0.00	47.40	0.514
mixed-c8	frequency	61.24	0.00	68.69	0.356
mixed-c8	lru	43.90	0.00	45.27	0.499
mixed-c8	static_state_adaptive	75.00	0.00	80.57	0.477
mixed-c8	full_0	81.26	0.00	97.50	0.422
mixed-c8	full_0.25	75.56	0.00	80.19	0.424
mixed-c8	full_0.5	73.87	0.00	85.37	0.244
mixed-c8	full_0.75	78.39	0.00	86.85	0.414
mixed-c8	full_1	80.82	0.00	87.98	0.520
mixed-c8	full_1.5	78.33	0.00	77.21	0.634
mixed-c8	full_2	68.69	0.00	55.92	0.652
rapid_high-c16	fixed_decay	85.92	0.00	88.59	0.749
rapid_high-c16	frequency	75.81	16.50	98.68	0.428
rapid_high-c16	lru	95.92	0.00	99.61	0.749
rapid_high-c16	static_state_adaptive	91.29	2.11	95.23	0.758
rapid_high-c16	full_0	76.92	19.44	100.00	0.497
rapid_high-c16	full_0.25	89.73	0.77	100.00	0.713
rapid_high-c16	full_0.5	92.66	0.14	100.00	0.712
rapid_high-c16	full_0.75	94.06	0.03	100.00	0.707
rapid_high-c16	full_1	93.40	0.02	99.87	0.719
rapid_high-c16	full_1.5	92.39	0.02	96.53	0.736
rapid_high-c16	full_2	90.44	0.02	89.96	0.743
rapid_high-c8	fixed_decay	45.10	0.00	46.74	0.473
rapid_high-c8	frequency	49.50	11.17	69.37	0.029
rapid_high-c8	lru	43.36	0.00	44.88	0.499
rapid_high-c8	static_state_adaptive	66.13	3.56	74.77	0.311
rapid_high-c8	full_0	66.63	14.79	97.87	0.158
rapid_high-c8	full_0.25	65.78	0.69	81.09	0.318
rapid_high-c8	full_0.5	70.80	0.13	85.65	0.214
rapid_high-c8	full_0.75	74.88	0.07	87.91	0.423
rapid_high-c8	full_1	76.28	0.10	89.19	0.472
rapid_high-c8	full_1.5	72.43	0.05	74.33	0.648
rapid_high-c8	full_2	63.99	0.06	55.82	0.669

Table 12: C2: all frozen operating points. Percentages except adaptation score.

Condition	Policy	Acc.	Stale	Ret.	Adapt.
mixed-c16	fixed_decay	95.27	0.00	98.83	0.748
mixed-c16	frequency	93.41	0.00	99.32	0.717
mixed-c16	lru	96.01	0.00	99.96	0.748
mixed-c16	static_state_adaptive	96.28	0.00	98.37	0.794
mixed-c16	full_0	83.33	0.00	100.00	0.422
mixed-c16	full_0.25	96.85	0.00	100.00	0.763
mixed-c16	full_0.5	96.81	0.00	100.00	0.769
mixed-c16	full_0.75	96.43	0.00	100.00	0.759
mixed-c16	full_1	96.20	0.00	100.00	0.754
mixed-c16	full_1.5	95.92	0.00	99.83	0.748
mixed-c16	full_2	95.96	0.00	99.87	0.748
mixed-c16	I_0	83.33	0.00	100.00	0.422
mixed-c16	I_0.25	96.85	0.00	100.00	0.763
mixed-c16	I_0.5	97.20	0.00	100.00	0.769
mixed-c16	I_0.75	97.58	0.00	100.00	0.784
mixed-c16	I_1	97.68	0.00	100.00	0.786

Condition	Policy	Acc.	Stale	Ret.	Adapt.
mixed-c16	I_1.5	97.89	0.00	100.00	0.793
mixed-c16	I_2	97.92	0.00	100.00	0.794
mixed-c16	C_0	83.33	0.00	100.00	0.422
mixed-c16	C_0.25	83.33	0.00	100.00	0.422
mixed-c16	C_0.5	83.33	0.00	100.00	0.422
mixed-c16	C_0.75	83.33	0.00	100.00	0.422
mixed-c16	C_1	83.33	0.00	100.00	0.422
mixed-c16	C_1.5	83.33	0.00	100.00	0.422
mixed-c16	C_2	83.33	0.00	100.00	0.422
mixed-c16	U_0	83.33	0.00	100.00	0.422
mixed-c16	U_0.25	83.33	0.00	100.00	0.422
mixed-c16	U_0.5	83.33	0.00	100.00	0.422
mixed-c16	U_0.75	83.33	0.00	100.00	0.422
mixed-c16	U_1	83.33	0.00	100.00	0.422
mixed-c16	U_1.5	83.33	0.00	100.00	0.422
mixed-c16	U_2	83.33	0.00	100.00	0.422
mixed-c8	fixed_decay	45.72	0.00	46.48	0.551
mixed-c8	frequency	61.96	0.00	71.77	0.362
mixed-c8	lru	43.97	0.00	45.62	0.531
mixed-c8	static_state_adaptive	74.39	0.00	79.44	0.473
mixed-c8	full_0	80.97	0.00	96.79	0.422
mixed-c8	full_0.25	74.65	0.00	78.52	0.422
mixed-c8	full_0.5	72.36	0.00	83.65	0.268
mixed-c8	full_0.75	78.31	0.00	86.63	0.428
mixed-c8	full_1	79.64	0.00	86.36	0.486
mixed-c8	full_1.5	76.62	0.00	73.80	0.624
mixed-c8	full_2	68.72	0.00	53.08	0.683
mixed-c8	I_0	80.97	0.00	96.79	0.422
mixed-c8	I_0.25	74.65	0.00	78.52	0.422
mixed-c8	I_0.5	73.42	0.00	77.31	0.422
mixed-c8	I_0.75	73.30	0.00	75.43	0.422
mixed-c8	I_1	73.47	0.00	75.39	0.422
mixed-c8	I_1.5	73.20	0.00	75.39	0.422
mixed-c8	I_2	73.23	0.00	75.39	0.422
mixed-c8	C_0	80.97	0.00	96.79	0.422
mixed-c8	C_0.25	80.97	0.00	96.79	0.422
mixed-c8	C_0.5	80.97	0.00	96.79	0.422
mixed-c8	C_0.75	80.97	0.00	96.79	0.422
mixed-c8	C_1	80.97	0.00	96.79	0.422
mixed-c8	C_1.5	80.97	0.00	96.79	0.422
mixed-c8	C_2	80.97	0.00	96.79	0.422
mixed-c8	U_0	80.97	0.00	96.79	0.422
mixed-c8	U_0.25	80.97	0.00	96.79	0.422
mixed-c8	U_0.5	80.97	0.00	96.79	0.422
mixed-c8	U_0.75	80.93	0.00	96.79	0.422
mixed-c8	U_1	80.93	0.00	96.79	0.422
mixed-c8	U_1.5	80.93	0.00	96.79	0.422
mixed-c8	U_2	80.96	0.00	96.79	0.423
rapid_high-c16	fixed_decay	85.85	0.00	88.59	0.748
rapid_high-c16	frequency	76.19	16.79	98.97	0.426
rapid_high-c16	lru	95.55	0.00	99.31	0.748
rapid_high-c16	static_state_adaptive	90.76	2.26	94.72	0.742
rapid_high-c16	full_0	77.11	18.99	100.00	0.538
rapid_high-c16	full_0.25	89.23	0.93	100.00	0.688
rapid_high-c16	full_0.5	92.13	0.24	100.00	0.707
rapid_high-c16	full_0.75	94.06	0.00	100.00	0.707
rapid_high-c16	full_1	92.89	0.03	99.57	0.712
rapid_high-c16	full_1.5	91.64	0.04	96.48	0.731
rapid_high-c16	full_2	90.44	0.00	90.91	0.736
rapid_high-c16	I_0	77.11	18.99	100.00	0.538
rapid_high-c16	I_0.25	81.90	12.64	100.00	0.688
rapid_high-c16	I_0.5	82.95	12.79	100.00	0.735
rapid_high-c16	I_0.75	83.20	13.00	100.00	0.755

Condition	Policy	Acc.	Stale	Ret.	Adapt.
rapid_high-c16	I_1	83.76	12.50	100.00	0.769
rapid_high-c16	I_1.5	84.21	12.41	100.00	0.779
rapid_high-c16	I_2	84.21	12.41	100.00	0.779
rapid_high-c16	C_0	77.11	18.99	100.00	0.538
rapid_high-c16	C_0.25	83.12	11.82	100.00	0.614
rapid_high-c16	C_0.5	83.37	11.78	100.00	0.609
rapid_high-c16	C_0.75	83.69	11.94	100.00	0.609
rapid_high-c16	C_1	83.70	11.94	100.00	0.609
rapid_high-c16	C_1.5	83.70	11.94	100.00	0.609
rapid_high-c16	C_2	83.70	11.94	100.00	0.609
rapid_high-c16	U_0	77.11	18.99	100.00	0.538
rapid_high-c16	U_0.25	77.27	18.80	100.00	0.538
rapid_high-c16	U_0.5	78.30	17.72	100.00	0.538
rapid_high-c16	U_0.75	80.16	15.79	100.00	0.553
rapid_high-c16	U_1	80.92	14.97	100.00	0.599
rapid_high-c16	U_1.5	81.89	13.98	100.00	0.624
rapid_high-c16	U_2	82.29	13.47	100.00	0.641
rapid_high-c8	fixed_decay	44.84	0.00	45.57	0.527
rapid_high-c8	frequency	50.29	11.47	71.44	0.105
rapid_high-c8	lru	43.14	0.00	44.92	0.500
rapid_high-c8	static_state_adaptive	65.23	4.01	72.47	0.295
rapid_high-c8	full_0	66.56	14.78	97.54	0.211
rapid_high-c8	full_0.25	65.89	0.58	80.14	0.336
rapid_high-c8	full_0.5	70.43	0.14	83.81	0.294
rapid_high-c8	full_0.75	73.47	0.04	87.15	0.395
rapid_high-c8	full_1	74.49	0.02	87.28	0.507
rapid_high-c8	full_1.5	71.61	0.05	72.66	0.623
rapid_high-c8	full_2	64.06	0.07	54.57	0.687
rapid_high-c8	I_0	66.56	14.78	97.54	0.211
rapid_high-c8	I_0.25	59.71	14.56	78.65	0.216
rapid_high-c8	I_0.5	59.37	14.53	77.82	0.255
rapid_high-c8	I_0.75	58.83	14.87	76.13	0.260
rapid_high-c8	I_1	58.34	15.11	75.64	0.260
rapid_high-c8	I_1.5	58.38	14.71	75.39	0.264
rapid_high-c8	I_2	58.38	14.71	75.39	0.281
rapid_high-c8	C_0	66.56	14.78	97.54	0.211
rapid_high-c8	C_0.25	67.04	13.73	97.54	0.211
rapid_high-c8	C_0.5	67.06	13.69	97.54	0.211
rapid_high-c8	C_0.75	67.10	13.66	97.54	0.211
rapid_high-c8	C_1	67.10	13.64	97.54	0.211
rapid_high-c8	C_1.5	67.10	13.64	97.54	0.211
rapid_high-c8	C_2	67.10	13.64	97.54	0.211
rapid_high-c8	U_0	66.56	14.78	97.54	0.211
rapid_high-c8	U_0.25	66.60	14.74	97.54	0.211
rapid_high-c8	U_0.5	66.91	14.24	97.54	0.211
rapid_high-c8	U_0.75	67.21	13.62	97.54	0.211
rapid_high-c8	U_1	67.40	13.20	97.54	0.211
rapid_high-c8	U_1.5	67.48	13.06	97.54	0.211
rapid_high-c8	U_2	67.49	13.04	97.54	0.211

Table 13: C3: all frozen operating points. Percentages except adaptation score.

Condition	Policy	Acc.	Stale	Ret.	Adapt.
contradictory_preference-c16	fixed_decay	84.65	4.40	88.77	0.857
contradictory_preference-c16	frequency	66.84	30.22	95.52	0.000
contradictory_preference-c16	adaptive_evolution	76.87	22.24	98.22	0.101
contradictory_preference-c8	fixed_decay	46.70	2.05	48.47	0.644
contradictory_preference-c8	frequency	37.75	16.60	56.27	0.000
contradictory_preference-c8	adaptive_evolution	64.89	16.65	71.81	0.022
preference_revival-c16	fixed_decay	83.78	0.00	88.77	0.782

Condition	Policy	Acc.	Stale	Ret.	Adapt.
preference_revival-c16	frequency	83.83	6.81	95.52	0.538
preference_revival-c16	adaptive_evolution	90.94	0.89	98.46	0.779
preference_revival-c8	fixed_decay	46.86	0.00	48.41	0.630
preference_revival-c8	frequency	46.29	4.12	56.27	0.257
preference_revival-c8	adaptive_evolution	71.68	0.00	72.68	0.485
preference_update-c16	fixed_decay	88.57	0.55	88.98	0.868
preference_update-c16	frequency	66.40	30.67	95.52	0.000
preference_update-c16	adaptive_evolution	73.39	25.72	98.22	0.000
preference_update-c8	fixed_decay	48.76	0.03	48.50	0.746
preference_update-c8	frequency	37.55	16.80	56.27	0.000
preference_update-c8	adaptive_evolution	62.58	18.87	70.32	0.013
temporary_task-c16	fixed_decay	85.27	0.22	88.77	0.839
temporary_task-c16	frequency	71.42	13.36	95.52	0.152
temporary_task-c16	adaptive_evolution	86.34	8.55	98.48	0.515
temporary_task-c8	fixed_decay	47.61	0.04	48.41	0.703
temporary_task-c8	frequency	40.01	6.11	56.27	0.013
temporary_task-c8	adaptive_evolution	71.64	6.92	73.35	0.362

Table 14: C4: all frozen operating points. Percentages except adaptation score.

Condition	Policy	Acc.	Stale	Ret.	Adapt.
fixed_cadence-c16-n1000	fixed_decay	80.19	0.00	88.26	0.490
fixed_cadence-c16-n1000	frequency	63.27	29.69	99.65	0.105
fixed_cadence-c16-n1000	lru	88.98	0.00	99.63	0.492
fixed_cadence-c16-n1000	static_state_adaptive	75.95	13.36	89.65	0.364
fixed_cadence-c16-n1000	full_0	68.01	27.21	99.97	0.188
fixed_cadence-c16-n1000	full_0.25	90.75	0.36	99.97	0.554
fixed_cadence-c16-n1000	full_1	87.81	0.00	99.97	0.467
fixed_cadence-c16-n1000	full_2	86.79	0.00	99.41	0.456
fixed_cadence-c16-n500	fixed_decay	82.03	0.00	89.50	0.490
fixed_cadence-c16-n500	frequency	73.21	19.09	99.62	0.243
fixed_cadence-c16-n500	lru	89.98	0.00	99.78	0.490
fixed_cadence-c16-n500	static_state_adaptive	89.21	3.55	95.20	0.591
fixed_cadence-c16-n500	full_0	79.21	18.00	99.93	0.475
fixed_cadence-c16-n500	full_0.25	90.44	0.26	99.95	0.513
fixed_cadence-c16-n500	full_1	88.85	0.00	99.95	0.475
fixed_cadence-c16-n500	full_2	87.68	0.00	99.59	0.458
fixed_cadence-c16-n5000	fixed_decay	79.37	0.00	87.52	0.498
fixed_cadence-c16-n5000	frequency	52.75	42.39	99.65	0.017
fixed_cadence-c16-n5000	lru	88.24	0.00	99.22	0.499
fixed_cadence-c16-n5000	static_state_adaptive	48.72	29.79	77.45	0.067
fixed_cadence-c16-n5000	full_0	60.57	26.75	99.99	0.050
fixed_cadence-c16-n5000	full_0.25	90.00	0.45	99.99	0.550
fixed_cadence-c16-n5000	full_1	87.19	0.00	99.99	0.474
fixed_cadence-c16-n5000	full_2	85.92	0.00	99.28	0.459
fixed_cadence-c8-n1000	fixed_decay	43.42	0.00	44.89	0.232
fixed_cadence-c8-n1000	frequency	40.74	19.03	69.39	0.019
fixed_cadence-c8-n1000	lru	42.03	0.00	43.97	0.210
fixed_cadence-c8-n1000	static_state_adaptive	48.07	12.73	67.29	0.056
fixed_cadence-c8-n1000	full_0	55.87	16.76	96.09	0.032
fixed_cadence-c8-n1000	full_0.25	66.90	0.55	92.26	0.159
fixed_cadence-c8-n1000	full_1	74.43	0.00	94.37	0.246
fixed_cadence-c8-n1000	full_2	60.19	0.00	55.91	0.374
fixed_cadence-c8-n500	fixed_decay	43.74	0.00	44.75	0.227
fixed_cadence-c8-n500	frequency	46.11	13.48	69.07	0.050
fixed_cadence-c8-n500	lru	42.78	0.00	44.97	0.206
fixed_cadence-c8-n500	static_state_adaptive	58.38	6.67	71.84	0.107
fixed_cadence-c8-n500	full_0	59.92	15.96	94.48	0.086
fixed_cadence-c8-n500	full_0.25	66.58	0.44	85.08	0.218
fixed_cadence-c8-n500	full_1	73.63	0.00	89.20	0.233

Condition	Policy	Acc.	Stale	Ret.	Adapt.
fixed_cadence-c8-n500	full_2	63.01	0.00	58.87	0.371
fixed_cadence-c8-n5000	fixed_decay	43.32	0.00	45.51	0.229
fixed_cadence-c8-n5000	frequency	36.07	26.23	69.52	0.003
fixed_cadence-c8-n5000	lru	41.57	0.00	44.37	0.202
fixed_cadence-c8-n5000	static_state_adaptive	32.77	20.54	57.31	0.009
fixed_cadence-c8-n5000	full_0	53.94	17.53	99.06	0.006
fixed_cadence-c8-n5000	full_0.25	67.39	0.61	98.47	0.108
fixed_cadence-c8-n5000	full_1	75.21	0.00	98.43	0.265
fixed_cadence-c8-n5000	full_2	57.95	0.00	52.77	0.399

H Reproducibility and Responsible Scope

The public repository is https://github.com/Nanoer-Wang/adaptive-memory-evolution. It includes the frozen heads, input hashes, public-data preparation, experiment runners, tests, and the archived summaries used here. The reproduction guide separates a small CPU replay from full sweeps and API-dependent language evaluation. A clean checkout can reconstruct the selected synthetic inputs without retraining.

The manuscript's `tools/build_assets.py` reads frozen JSON summaries and creates every new table and vector plot. Source hashes are recorded in `evidence/source_manifest.json`. It never trains or re-estimates model parameters. Experiment entry points in the original project are:

- `scripts/validation_matrix.py`: structured validation matrix.
- `scripts/validation_language.py`: conversational transfer.
- `scripts/characterization.py`: rate control and component flow interventions.
- `scripts/characterization_semantic.py`: semantic migration.
- `scripts/characterization_stress.py`: long-horizon stress.

Existing completed outputs are verified before reuse. The manuscript package includes summarized evidence and source paths; multi-gigabyte event archives remain indexed in the project rather than duplicated into the paper ZIP.

Public benchmark text, generated schedules, and released model weights are used; no new human-subject study is claimed. Personal-memory deployment can create privacy and deletion risks even when a benchmark is public. Consistency scores do not establish factual truth, safe deletion, or reliable handling of retractions. A practical system should expose record provenance and distinguish removing a resident copy from deleting every persistent trace. These considerations motivate caution in transferring a memory-management result to a user-facing assistant, rather than claims that this paper has solved those issues.

I Relationship to Published Memory Systems

We inspected the primary papers for A-MEM and Mem0 and the official LongMemEval and LoCoMo evaluation instructions. This is a literature-based protocol comparison, not a reproduction. A shared dataset name is insufficient for comparing performance: question IDs, task categories, reader and judge, prompts, retrieval budget, storage limits, and available feedback must also agree.

Table 15: Mechanism and protocol comparison. No row establishes a cross-paper performance ranking.

Work	What changes in memory	Evaluation compatibility with this study
A-MEM [4]	New notes can revise historical attributes and links through LLM-based operations.	LoCoMo QA with multiple readers; the inspected version reports F1/BLEU-1 and category-dependent retrieval settings. Our 24-question subset, reader, scoring, and 64-slot limit differ.
Mem0 [6]	LLM-mediated addition, updating, deletion, or no operation; a graph variant tracks relations.	LoCoMo evaluation excludes adversarial questions and uses GPT-4o-mini operations plus judge-based scores. The question set, scoring pipeline, and budget differ from ours.
AME	Immutable versioned content plus four scalar states; frozen weights and explicit rate interventions control residency.	Structured shared-stream experiments isolate interventions; language transfer uses Agnes on fixed 24-question subsets. These are not official full-benchmark results.

LongMemEval's official release contains 500 instances per variant and provides a GPT-4o evaluation entry point [13]; our transfer test uses a balanced 24-question subset and a same-backend Agnes judge. LoCoMo's public release and scoring pipeline likewise differ from our custom transfer protocol [14]. Published scores are therefore not pooled with our results, and we make no cross-paper performance claim. The supported distinction is the controlled study of retention–adaptation operating points, not superiority over these complete memory systems. A matched comparison remains future work.